Chapter5

Light Ray (Classwork)

1. A pinhole camera of length 20 cm is used to view the image of a tree of height 12 m which is 40 m away from the pinhole. Calculate the height of the image of the tree obtained on the screen.

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2. If the pinhole camera, in Example1 is moved by 10 m towards the tree, what will be the height of the tree on the screen?

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3. A pinhole camera is used to take the photograph of a person who is 4 m away from the pinhole. If the length of the box used is 18 cm and the height of the image of the person is 9 cm, calculate the:
(a) magnification produced by the pinhole camera.
(b) height of the person.

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4. A tree 18 m high is observed with a pinhole camera that is placed 40 m away. If the camera is 20 cm long, find the height of the image formed.

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5. The length of a pinhole camera is 10 cm. It forms an image of linear magnification of 0.2. Find the position of the object.

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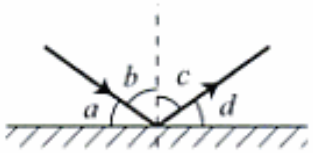
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6. The figure below shows a light ray reflecting off a smooth surface.



A-The angle of incidence is …….
B-The angle of reflection is …….
C-If a=42°, what is the angle of reflection?

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7. A light ray strikes a mirror at an angle of 72° to the normal line.
A-What is the angle of reflection?
B-What is the angle between the incident ray and reflected ray?

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8. A ray strikes a mirror at angle of 26° to the mirror surface. What is the angle between the incident ray and the reflected ray?

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9. An incident ray strikes a smooth reflective surface at an angle of 33° to the surface normal. What is the angle of reflection?

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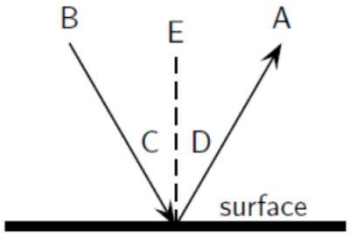
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10. In the diagram, label the following:

(a) normal ……
(b) angle of incidence ……
(c) angle of reflection ……
(d) incident ray ……
(e) reflected ray ……



11. A ray of light strikes a surface at 15◦ to the surface normal. Draw a ray diagram showing the incident ray, reflected ray and surface normal. Calculate the angles of incidence and reflection and fill them in on your diagram.

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12. A ray of light strikes a mirror at an angle of 38° to the normal. What is the angle that the reflected angle makes with the normal?

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13. Figure 12.5 shows a ray of light incident on a mirror.
(a) State the relationship between the angle of incidence and the angle of reflection.
(b) Complete the diagram to show the reflected ray.
(c) What is the angle of incidence?
(d) What is the angle of reflection?

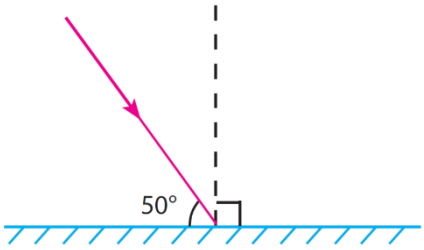


Figure 12.5

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14. Figure 13.11 shows how a periscope uses mirrors to allow the user to see beyond obstacles.
a. Copy and complete the diagram in Figure 13.11 to show how the light reaches the observer.
b. The image the observer sees is not laterally inverted. Explain why.

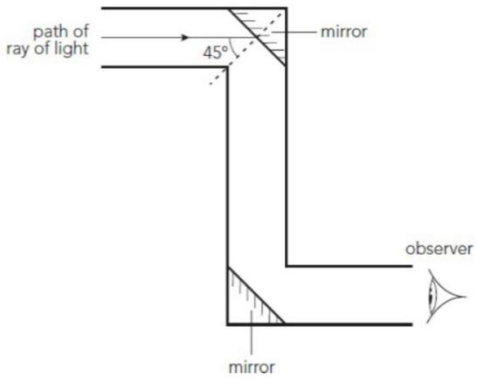


Figure 13.11: Periscope.

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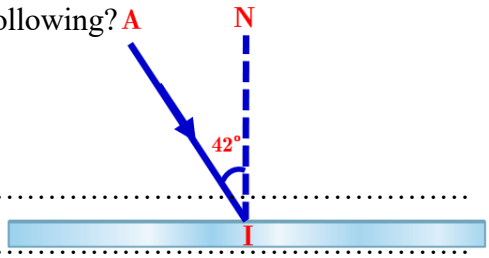
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15. If the angle of incidence of a ray of light is 42°, what is each of the following?

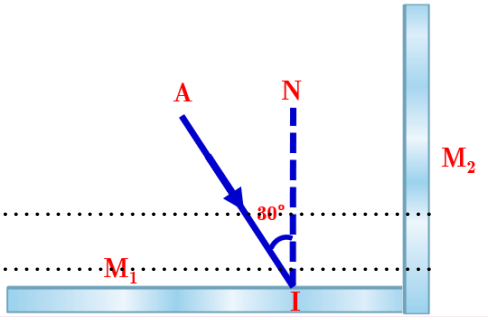


a. the angle of reflection

b. the angle the incident ray makes with the mirror

c. the angle between the incident ray and the reflected ray

16. Two plane mirrors are positioned at right angles to one another.
A ray of light strikes one mirror at an angle of 30° to the normal.
It then reflects toward the second mirror.
What is the angle of reflection of the light ray off the second mirror?



17. Figure 27.6 shows a ray of light PQ striking a mirror AB. The mirror AB and the mirror CD are at right angles to each other. QN is a normal to the mirror AB.

a. What is the value of the angle of incidence of the ray PQ on the mirror AB?

b. Copy the diagram, and continue the ray PQ to show the path it takes after reflection at both mirrors.

c. What are the values of the angle of reflection at AB, the angle of incidence at CD and the angle of reflection at CD?

d. What do you notice about the path of the ray PQ and the final reflected ray?

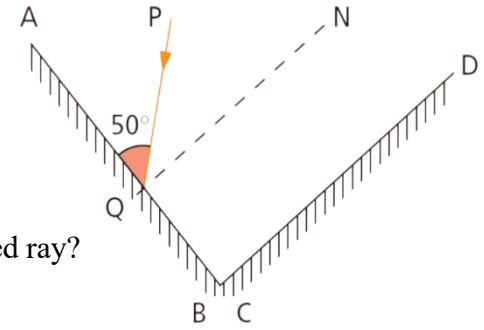
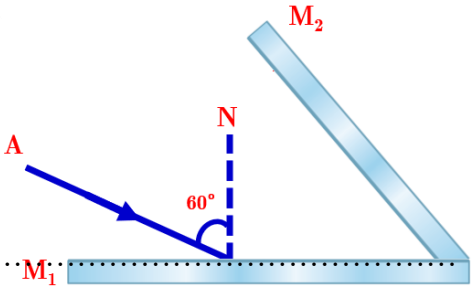


Figure 27.6

18. A ray of light strikes a plane mirror at an angle of incidence of 60°, is reflected from the mirror and then strikes a second plane mirror placed so that the angle between the mirrors is 45°.
The angle of reflection at the second mirror, in degrees, is
A 15 B 25 C 45 D 65 E 75



19. A person stands in front of a mirror (Figure 27.7). How much of the mirror is used to see from eye to toes?

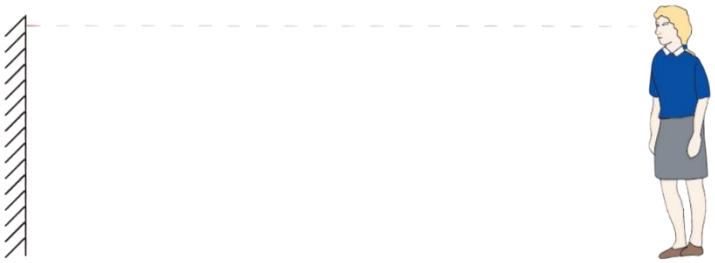


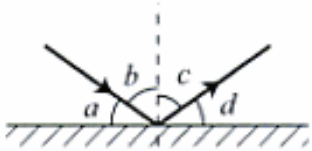
Figure 27.7

Chapter4 Light

1. The length of a pinhole camera is 10 cm. It forms an image of linear magnification of 0.2. Find the position of the object.

2. State three characteristics of the image formed by a pinhole camera.

3. The figure below shows a light ray reflecting off a smooth surface.



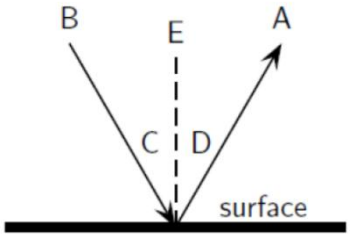
A-The angle of incidence is …….
B-The angle of reflection is …….
C-If a=42°, what is the angle of reflection?

4. A light ray strikes a mirror at an angle of 72° to the normal line.
A-What is the angle of reflection?
B-What is the angle between the incident ray and reflected ray?

5. A ray strikes a mirror at angle of 26° to the mirror surface. What is the angle between the incident ray and the reflected ray?

6. An incident ray strikes a smooth reflective surface at an angle of 33° to the surface normal. What is the angle of reflection?

7. In the diagram, label the following:



(a) normal ……
(b) angle of incidence ……
(c) angle of reflection ……
(d) incident ray ……
(e) reflected ray ……

8. A ray of light strikes a surface at 15◦ to the surface normal. Draw a ray diagram showing the incident ray, reflected ray and surface normal. Calculate the angles of incidence and reflection and fill them in on your diagram.

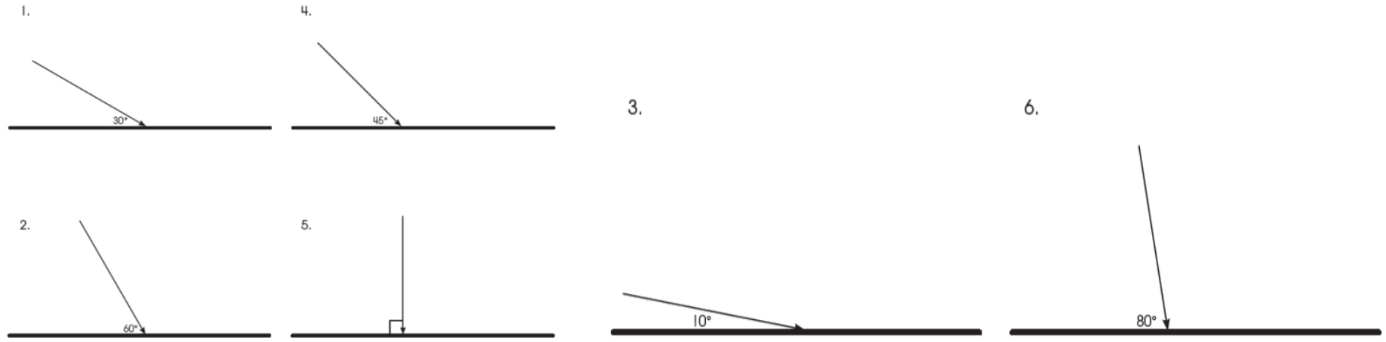
9. A ray of light strikes a mirror at an angle of 38° to the normal. What is the angle that the reflected angle makes with the normal?

10. A ray of light strikes a mirror at an angle of 53° to the normal.
a. What is the angle of reflection?
b. What is the angle between the incident ray and the reflected ray?

11. If the angle of incidence of a ray of light is 42°, what is each of the following?
a. the angle of reflection
b. the angle the incident ray makes with the mirror
c. the angle between the incident ray and the reflected ray

12. A student does an experiment with a mirror. She measures the angle between the *mirror* and an incident ray and finds that it is 30°.
a. What is the angle of incidence?
b. What is the angle of reflection?

13. Draw the expected path of the light ray as it reflects off each mirror.



14. A ray of light incident upon a mirror makes an angle of 36° with the mirror. What is the angle between the incident ray and the reflected ray?

15. If a light ray reflects off a plane mirror at an angle of 35° to the normal, what was the angle of incidence of the ray?

16. Light from a laser strikes a plane mirror at an angle of 38° to the normal. If the laser is moved so that the angle of incidence increases by 13°, what is the new angle of reflection?

17. Two plane mirrors are positioned at right angles to one another. A ray of light strikes one mirror at an angle of 30° to the normal. It then reflects toward the second mirror. What is the angle of reflection of the light ray off the second mirror?

18. Two adjacent plane mirrors form a right angle, as shown in **Figure 17-19.** A light ray is incident upon one of the mirrors at an angle of 30° to the normal.

a. What is the angle at which the light ray is reflected from the other mirror?

b. A retroreflector is a device that reflects incoming light rays back in a direction opposite to that of the incident rays. Draw a diagram showing the angle of incidence on the first mirror for which the mirror system acts as a retroreflector.

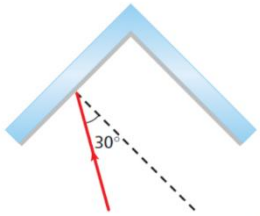


Figure 17-19

19. Draw a ray diagram of a plane mirror to show that if you want to see yourself from your feet to the top of your head, the mirror must be at least half your height.

20. Two plane mirrors are connected at their sides so that they form a 45° angle between them. A light ray strikes one mirror at an angle of 30° to the normal and then reflects off the second mirror. Calculate the angle of reflection of the light ray off the second mirror.

Chapter4 Mirrors (concave and convex mirror)

1. Summary Equation

Mirror Equation : $\frac{1}{f}=\frac{1}{d_o}+\frac{1}{d_i}\Rightarrow f=\frac{d_o d_i}{d_o+d_i}$

Object's distance : $\frac{1}{f}=\frac{1}{d_o}+\frac{1}{d_i}\Rightarrow \frac{1}{d_o}=\frac{1}{f}-\frac{1}{d_i}\Rightarrow d_o=\frac{fd_i}{d_i-f}$

Image's distance : $\frac{1}{f}=\frac{1}{d_o}+\frac{1}{d_i}\Rightarrow \frac{1}{d_i}=\frac{1}{f}-\frac{1}{d_o}\Rightarrow d_i=\frac{fd_o}{d_o-f}$

Magnification : $m=-\frac{d_i}{d_o}=\frac{h_i}{h_o}$

➲ **Units**

- $f = R/2$ is the focal length measured in (m) or (cm).
- d_o is the object's distance measured in (m) or (cm).
- d_i is the image's distance measured in (m) or (cm).
- h_o is the object's height measured in (m) or (cm).
- h_i is the image's height measured in (m) or (cm).
- R is the radius of curvature measured in (m) or (cm).
- m is the magnification (unitless).

2. A concave mirror has a focal length of 10.0 cm. What is its radius of curvature?

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3. An object is placed 50.0 cm from a concave spherical mirror with focal length of magnitude 20.0 cm. (a) Find the location of the image. (b) What is the magnification of the image? (c) Is the image real or virtual? (d) Is the image upright or inverted?

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4. A concave spherical mirror has a radius of curvature of magnitude 20.0 cm. (a) Find the location of the image for object distances of (i) 40.0 cm, (ii) 20.0 cm, and (iii) 10.0 cm. For each case, state whether the image is (b) real or virtual and (c) upright or inverted. (d) Find the magnification in each case.

5. An object is placed 20.0 cm from a concave spherical mirror having a focal length of magnitude 40.0 cm. (a) Use graph paper to construct an accurate ray diagram for this situation. (b) From your ray diagram, determine the location of the image. (c) What is the magnification of the image? (d) Check your answers to parts (b) and (c) using the mirror equation.

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6. Describe the image produced by the object in Figure 17-21 as real or virtual, inverted or upright, and smaller or larger than the object.

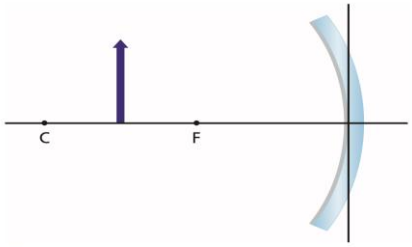


Figure 17-21

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7. Star Image Light from a star is collected by a concave mirror. How far from the mirror is the image of the star if the radius of curvature is 150 cm?

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8. Find the image position and height for the object shown in Figure 17-22.

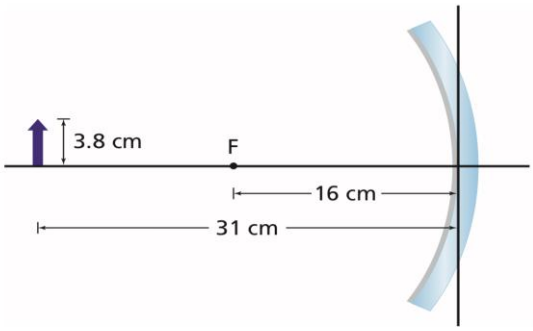


Figure 17-22

9. A convex spherical mirror has a radius of curvature of magnitude 40.0 cm. Determine the position of the virtual image and the magnification for object distances of (a) 30.0 cm and (b) 60.0 cm. (c) Are the images in parts (a) and (b) upright or inverted?

10. An object is 30.0 cm from a concave mirror of 15.0 cm focal length. The object is 1.8 cm tall. Use the mirror equation to find the image position. What is the image height?

11. Dental Mirror A dentist uses a small mirror with a radius of 40 mm to locate a cavity in a patient's tooth. If the mirror is concave and is help 16 mm from the tooth, what is the magnification of the image?

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12. A 3.0-cm-tall object is 22.4 cm from a concave mirror. If the mirror has a radius of curvature of 34.0 cm, what are the image position and height?

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13. **Jeweler's Mirror** A jeweler inspects a watch with a diameter of 3.0 cm by placing it 8.0 cm in front of a concave mirror of 12.0-cm focal length.

a. Where will the image of the watch appear?

b. What will be the diameter of the image?

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14. Sunlight falls on a concave mirror and forms an image that is 3.0 cm from the mirror. An object that is 24 mm tall is placed 12.0 cm from the mirror.

a. Sketch the ray diagram to show the location of the image.

b. Use the mirror equation to calculate the image position.

c. How tall is the image?

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Chapter4 (Refraction of light)

I. Refraction of light

1. A thick sheet of plastic, n = 1.500, is used as the side of an aquarium tank. Light reflected from a fish in the water has an angle of incidence of 35.0°. At what angle does the light enter the air?
   Given: Refractive index of water is 1.33, $\sin 35.0^\circ = 0.57$, $\sin^{-1}(0.509) = 30.6^\circ$, $\sin 30.6^\circ = 0.51$, $\sin^{-1}(0.76) = 49.9^\circ$.
2. A diamond's index of refraction for red light, 656 nm, is 2.410, while that for blue light, 434 nm, is 2.450. Suppose that white light is incident on the diamond at 30.0°. Find the angles of refraction for red and blue light.
   Given: $\sin 30.0^\circ = 0.5$, $\sin^{-1}(0.2075) = 12.0^\circ$, $\sin^{-1}(0.2041) = 11.8^\circ$.
3. The index of refraction of crown glass is 1.53 for violet light, and it is 1.51 for red light.
   a. What is the speed of violet light in crown glass?
   b. What is the speed of red light in crown glass?
   Given: Speed of light in air is 3×10^8 m/s.
4. A ray of light in a tank of water has an angle of incidence of 55.0°. What is the angle of refraction in air?
   Given: Refractive index of water is 1.33, $\sin 55.0^\circ = 0.82$, $\sin^{-1}(1.089) =$ error.
5. The ray of light shown in Figure 18-27 is incident upon a 60°-60°-60° glass prism, n= 1.5.
   a. Using Snell's law of refraction, determine the angle, θ_2, to the nearest degree.
   b. Using elementary geometry, determine the value of θ_1'.
   c. Determine θ_2'.

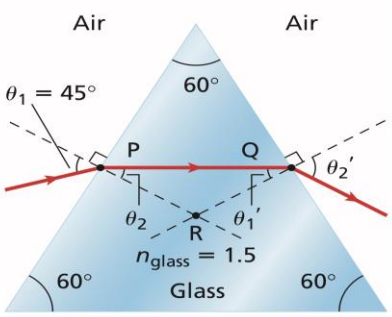


Figure 18-27

6. The speed of light in a clear plastic is 1.90×10^8 m/s. A ray of light strikes the plastic at an angle of 22.0°. At what angle is the ray refracted?

II. Lenses

7. An object is placed 32.0 cm from a convex lens that has a focal length of 8.0 cm.
   a. Where is the image?
   b. If the object is 3.0 cm high, how tall is the image?
   c. What is the orientation of the image?
8. A 2.25-cm-tall object is 8.5 cm to the left of a convex lens of 5.5-cm focal length. Find the image position and height.
9. An object near a convex lens produces a 1.8-cm-tall real image that is 10.4 cm from the lens and inverted. If the focal length of the lens is 6.8 cm, what are the object position and height?
10. An object is placed to the left of a convex lens with a 25-mm focal length so that its image is the same size as the object. What are the image and object positions?
11. Use a scale ray diagram to find the image position of an object that is 30 cm to the left of a convex lens with a 10-cm focal length.
12. Calculate the image position and height of a 2.0-cm-tall object located 25 cm from a convex lens with a focal length of 5.0 cm. What is the orientation of the image?
13. A newspaper is held 6.0 cm from a convex lens of 20.0-cm focal length. Find the image position of the newsprint image.

14. An object is placed 32.0 cm from a convex lens that has a focal length of 8.0 cm.
a. Where is the image?
b. If the object is 3.0 cm high, how tall is the image?
c. What is the orientation of the image?

15. A 2.25-cm-tall object is 8.5 cm to the left of a convex lens of 5.5-cm focal length. Find the image position and height.

16. An object near a convex lens produces a 1.8-cm-tall real image that is 10.4 cm from the lens and inverted. If the focal length of the lens is 6.8 cm, what are the object position and height?

17. An object is placed to the left of a convex lens with a 25-mm focal length so that its image is the same size as the object. What are the image and object positions?

18. Use a scale ray diagram to find the image position of an object that is 30 cm to the left of a convex lens with a 10-cm focal length.

19. Calculate the image position and height of a 2.0-cm-tall object located 25 cm from a convex lens with a focal length of 5.0 cm. What is the orientation of the image?

20. A newspaper is held 6.0 cm from a convex lens of 20.0-cm focal length. Find the image position of the newsprint image.

21. A magnifying glass has a focal length of 12.0 cm. A coin, 2.0 cm in diameter, is placed 3.4 cm from the lens. Locate the image of the coin. What is the diameter of the image?

22. A convex lens with a focal length of 22.0 cm is used to view a 15.0-cm-long pencil located 10.0 cm away. Find the height and orientation of the image.

23. A stamp collector wants to magnify a stamp by 4.0 when the stamp is 3.5 cm from the lens. What focal length is needed for the lens?

24. A magnifier with a focal length of 30 cm is used to view a 1-cm-tall object. Use ray tracing to determine the location and size of the image when the magnifier is positioned 10 cm from the object.

25. A converging lens has a focal length of 25.5 cm. If it is placed 72.5 cm from an object, at what distance from the lens will the image be?

26. If an object is 10.0 cm from a converging lens that has a focal length of 5.00 cm, how far from the lens will the image be?

27. A convex lens is needed to produce an image that is 0.75 times the size of the object and located 24 cm from the lens on the other side. What focal length should be specified?

28. An object is located 14.0 cm from a convex lens that has a focal length of 6.0 cm. The object is 2.4 cm high.
a. Draw a ray diagram to determine the location, size, and orientation of the image.
b. Solve the problem mathematically.

29. A 3.0-cm-tall object is placed 22 cm in front of a converging lens. A real image is formed 11 cm from the lens. What is the size of the image?

30. A 3.0-cm-tall object is placed 15.0 cm in front of a converging lens. A real image is formed 10.0 cm from the lens.
a. What is the focal length of the lens?
b. If the original lens is replaced with a lens having twice the focal length, what are the image position, size, and orientation?

31. A diverging lens has a focal length of 15.0 cm. An object placed near it forms a 2.0-cm-high image at a distance of 5.0 cm from the lens.
a. What are the object position and object height?
b. The diverging lens is now replaced by a converging lens with the same focal length. What are the image position, height, and orientation? Is it a virtual image or a real image?